



SCOPE - X

Electronics at Home

User Manual

Complete Laboratory Manual

- Bipolar Supply $\pm 12V$ added • Offset and Gain adjust simplified • Additional breadboard space

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1 Introduction

Overview: The Scope-X development board from Bumblebee Instruments provides a comprehensive platform for understanding basic analog circuit design principles through hands-on experimentation and analysis.

This manual serves as both an instructional guide and reference document for the Scope-X. Designed by Bumblebee Instruments, the kit provides students and engineers with practical experience in analog circuit design, analysis, and implementation using industry-standard components and methodologies.

The laboratory exercises contained within this manual progress from fundamental operational amplifier concepts to advanced signal processing techniques, ensuring a thorough understanding of analog system design principles.

1.1 Learning Objectives

Upon completion of this manual, users will be able to:

- Design and analyze operational amplifier circuits
- Implement active filter topologies
- Understand analog signal processing fundamental
- Troubleshoot analog circuit implementations

1.2 Kit Components

Table 1: Hardware Components Overview

Component	Description	Quantity
Scope-X Main Board	Analog signal acquisition and processing unit	1
Power Cable	Supplies regulated power to the main board	1
USB Interface Cable	Enables communication with the PC	1

2 System Setup and Configuration

Important: Ensure all power connections are verified before energizing the system. Observe proper ESD handling procedures when working with electronic components.

2.1 Setup Procedure

Prerequisites:

- A computer with a valid USB port.
- The Scope-X Software Suite and Drivers (available for download).
- Administrator access for driver installation.

Quick Start Guide

1. **Software Installation:** Install the Scope-X application and the requisite USB drivers for your specific operating system (Windows, Linux, or macOS) from the provided resource folder.
2. **Hardware Connection:** Connect the Scope-X board to your computer using the USB cable. Connect the power supply to the board.
3. **Device Recognition:** Allow your operating system to initialize the device. It should appear as a USB Serial/COM device.
4. **Launch:** Open the Scope-X GUI. Use the 'Check Ports' or 'Connect' feature to establish communication with the board.

Connection Status

When successfully connected, the application will display a "Ready" or "Connected" status indicator.

3 Hardware Architecture

3.1 Main Board Overview

The Scope-X main board features a modular architecture optimized for educational use and rapid prototyping.

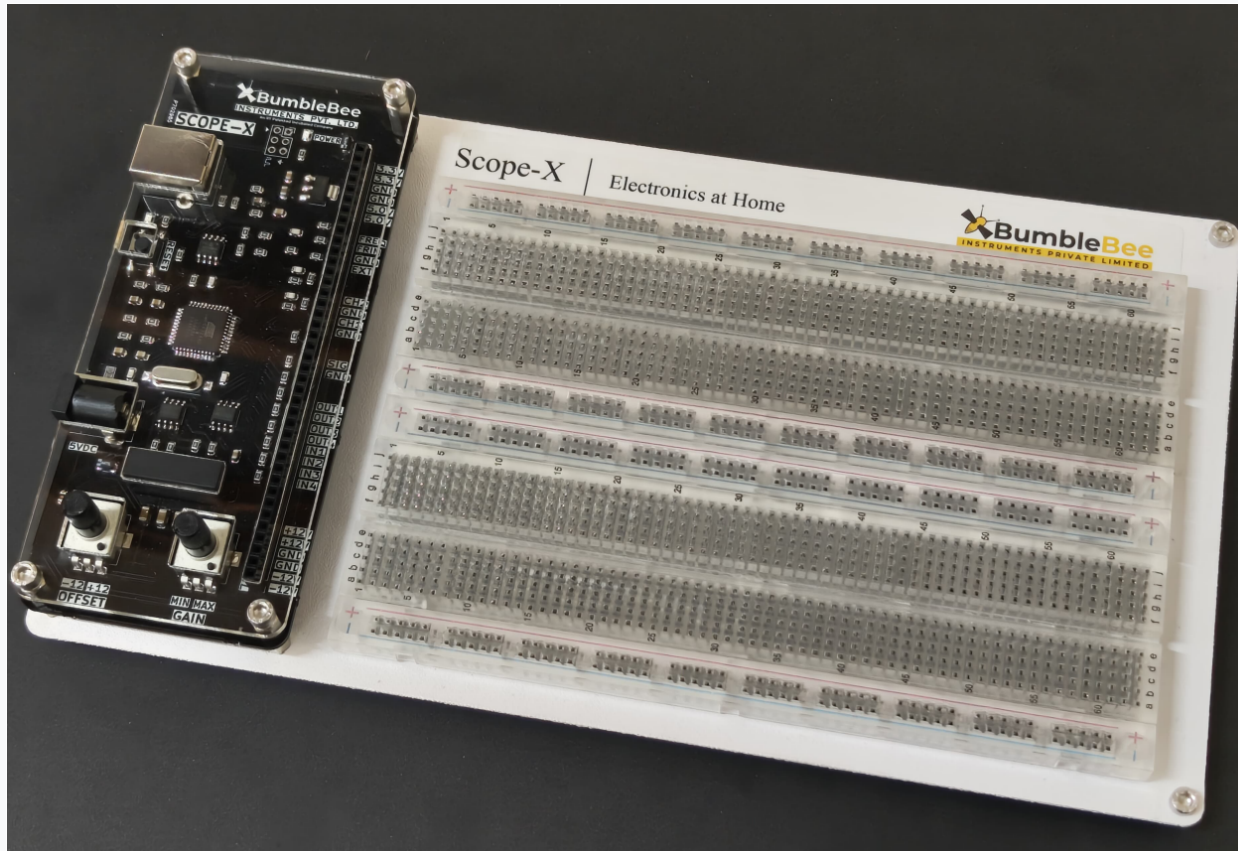


Figure 1: Scope-X Main Board Layout

3.2 Pinout and Features

Table 2: Board Functions

Label	Function / Usage
USB-B	PC Interface Connection
Power	Main Power Input
Reset	System Reset Button
External	Auxiliary External Pins
Scope	Oscilloscope Channels (1 & 2)
Signal Gen	Function Generator Output
5VDC	On-board Logic Supply
Digital I/O	4-bit Logic Input/Output
Offset	Signal Offset Adjustment (-12V to +12V)
Gain	Signal Gain Adjustment
Dual Rail	Bipolar Power Supply Output

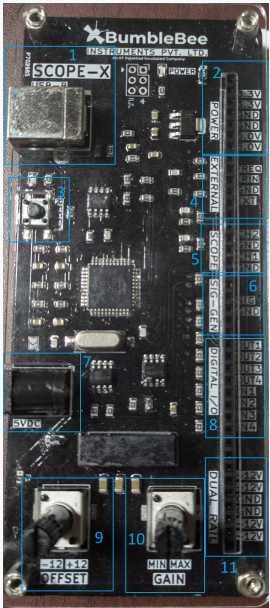
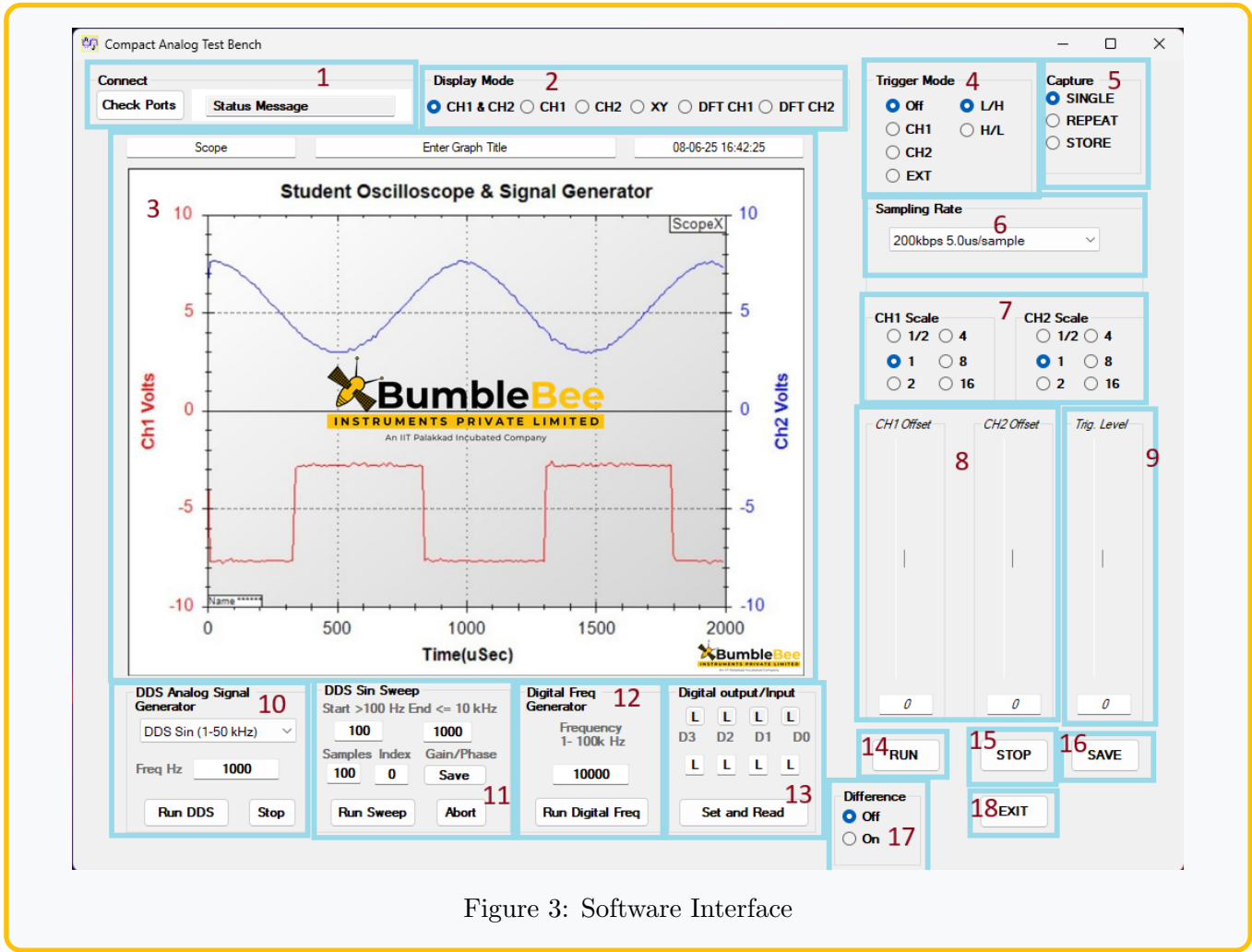


Figure 2: Board Detail View

4 GUI Software Reference

The Scope-X GUI provides a centralized interface for signal generation, acquisition, and analysis. (VB.NET version)



bumbleyellowID	Function Description
1	Connect/Port Check: Scans available COM ports and establishes connection to the board. Indicates "Ready" status upon success.
2	Display Mode Selector: Toggles visualization modes (CH1 & CH2, Single Channel, X-Y Plot, or DFT/FFT views).
3	Main Display Area: Real-time visualization of waveforms. Hovering provides point values. Right-click accesses Save/Print/Zoom context menu.
4	Trigger Control: Configures trigger source (CH1, CH2, EXT) and slope (Rising/Falling Edge).
5	Capture Mode: Selects acquisition method: <i>Single</i> (one-shot), <i>Repeat</i> (continuous), or <i>Store</i> (overlay traces).
6	Timebase/Sampling: Adjusts the sampling rate and horizontal time scale (e.g., $\mu\text{s}/\text{sample}$).
7	Vertical Scale (Gain): Sets input gain/attenuation for CH1 and CH2 to optimize signal visibility.
8	Offset Control: Adjusts vertical position to center signals with DC offsets.
9	Trigger Level: Sets the voltage threshold for the trigger event.
10	Analog Signal Generator: Controls the DDS output. Select frequency, waveform type (Sine, Square, etc.), or load Arbitrary Waveforms (.csv).
11	Bode Plot / Sweep: Performs frequency sweep analysis (Start/End Freq) to generate and save Amplitude/Phase response data.
12	Digital Frequency Gen: Generates square waves (Normal/Inverted) at specified frequencies on digital pins.
13	Digital I/O Control: Manually toggle output bits (D0-D3) and read input states.
14	RUN: Initiates data acquisition.
15	STOP: Halts data acquisition.
16	SAVE: Exports current buffer data (Time, CH1, CH2, FFT data) to .csv format.
17	Math Functions: Displays calculated waveforms (e.g., Difference: CH1 - CH2).
18	EXIT: Closes the application.

5 Experiment 1: Inverting Amplifier

Objective: Design an amplifier with a Gain of -10 using IC 741.

Theory and Design

The inverting amplifier provides a phase-shifted output (180°) amplified by the ratio of the feedback resistor to the input resistor.

Formula:

$$V_{out} = - \left(\frac{R_f}{R_{in}} \right) V_{in} \quad (1)$$

Design Parameters:

- **Target Gain (A_v):** -10
- **Let R_{in} :** 1 k Ω
- **Calculate R_f :**

$$R_f = |A_v| \times R_{in} = 10 \times 1k = 10 \text{ k}\Omega$$

Procedure:

1. Connect Op-Amp Pin 2 to R_{in} .
2. Connect R_f between Pin 2 and Pin 6.
3. Ground Pin 3 (Non-Inverting Input).

Circuit Diagram

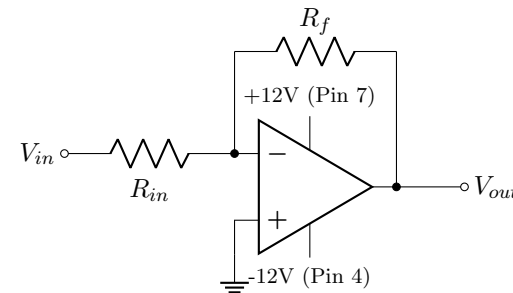


Figure 4: Inverting Amplifier Connection

6 Experiment 2: Non-Inverting Amplifier

Objective: Design an amplifier with a Gain of 5 using IC 741.

Theory and Design

The non-inverting amplifier provides amplification without phase inversion. The input signal is applied directly to the non-inverting terminal.

Formula:

$$V_{out} = \left(1 + \frac{R_f}{R_1}\right) V_{in} \quad (2)$$

Design Parameters:

- **Target Gain (A_v):** 5
- **Let R_1 :** 1 k Ω
- **Calculate R_f :**

$$5 = 1 + \frac{R_f}{1k} \implies \frac{R_f}{1k} = 4$$

$$R_f = 4 \text{ k}\Omega$$

Procedure:

1. Apply V_{in} to Pin 3.
2. Connect voltage divider (R_1, R_f) to Pin 2.
3. Ground the other end of R_1 .

Circuit Diagram

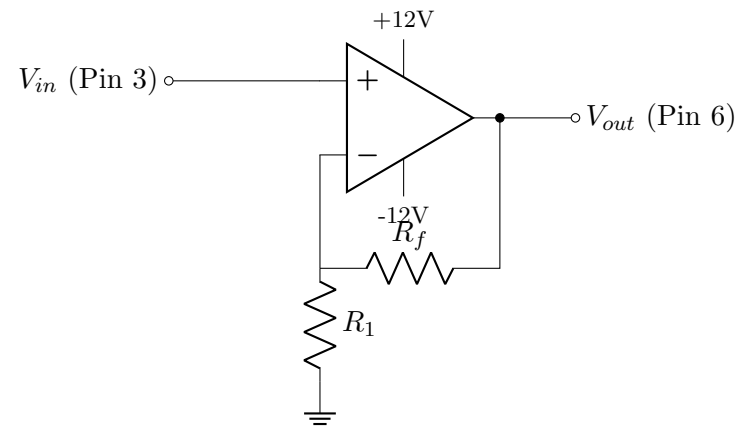


Figure 5: Non-Inverting Amplifier Connection

7 Experiment 3: Active Low-Pass Filter

Objective: Design a filter to attenuate frequencies above 1 kHz.

Theory and Design

An active Low-Pass Filter (LPF) allows low frequency signals to pass but blocks high frequencies. By placing a capacitor in parallel with the feedback resistor, the gain decreases as frequency increases.

Formula (Cut-off Frequency):

$$f_c = \frac{1}{2\pi R_f C_f} \quad (3)$$

Design Parameters:

- **Target f_c :** ≈ 1 kHz
- **Let $R_f = R_{in}$:** 10 k Ω (Unity DC Gain)
- **Calculate C_f :**

$$C_f = \frac{1}{2\pi(10k)(1k)} \approx 15.9 \text{ nF}$$

Use standard value: 15 nF or 16 nF.

Circuit Diagram

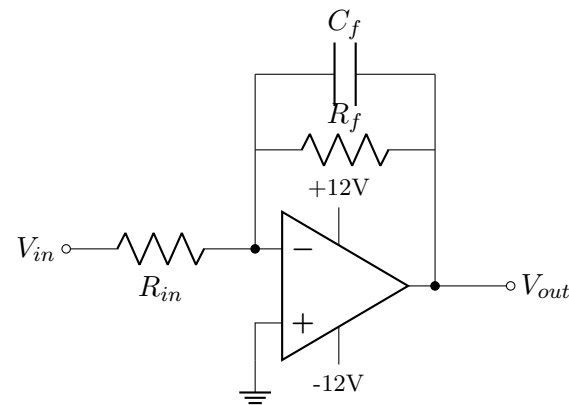


Figure 6: Active LPF (Inverting Topology)

8 Experiment 4: Active High-Pass Filter

Objective: Design a filter to block DC and low frequencies (Cut-off 100 Hz).

Theory and Design

An active High-Pass Filter (HPF) blocks low frequencies. This is achieved by placing a capacitor in series with the input resistor. At low frequencies, the capacitor impedance is high, blocking the signal.

Formula:

$$f_c = \frac{1}{2\pi R_{in} C_{in}} \tag{4}$$

Design Parameters:

- Target f_c : ≈ 100 Hz
- Let R_{in} : 10 k Ω
- Calculate C_{in} :

$$C_{in} = \frac{1}{2\pi(10k)(100)} \approx 159 \text{ nF}$$

Use standard value: 160 nF or 0.16 μ F.

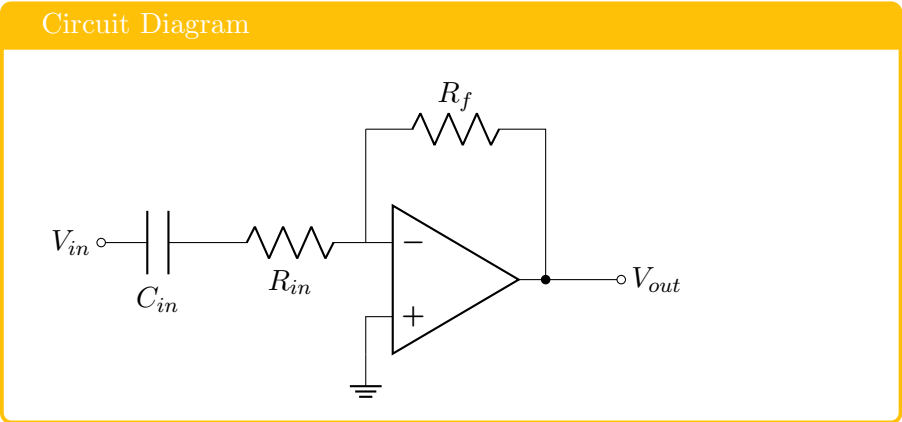


Figure 7: Active HPF (Inverting Topology)

Troubleshooting Guide

Symptom	Solution
Clipping Output	Input signal is too large. Reduce input amplitude or gain.
No Signal (Flatline)	Check power rails ($\pm 12V$) and breadboard connectivity.
50Hz/60Hz Noise	Ensure the board Ground (GND) is connected to the circuit ground.